

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: To Study the basics of Dart language and design a basic Flutter App to Create Unit convertor(for length, weight, temperature, Area) app using Flutter.	
Experiment No:- 02	Date:- 20-09-2024	Enrollment No:- 92200133030	

Objective:- To Study the basics of Dart language and design a basic Flutter App to Create Unit convertor(for length, weight, temperature, Area) app using Flutter.

Code :-

```
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        useMaterial3: true,
      ),
      home: ConverterScreen(),
    );
  }
}

class ConverterScreen extends StatefulWidget {
  @override
  _ConverterScreenState createState() => _ConverterScreenState();
}

class _ConverterScreenState extends State<ConverterScreen> {
  // Conversion options with units (from -> to)
  final List<String> _conversionTypes = [
    'Meters to Kilometers',
    'Kilometers to Meters',
    'Grams to Kilograms',
    'Kilograms to Grams',
    'Celsius to Fahrenheit',
    'Fahrenheit to Celsius',
    'Square Meters to Hectares',
    'Hectares to Square Meters',
    // New conversions
    'Miles to Kilometers',
    'Kilometers to Miles',
    'Pounds to Kilograms',
    'Kilograms to Pounds',
    'Liters to Milliliters',
  ];
```

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```

'Milliliters to Liters',
'Kilometers per hour to Miles per hour',
'Miles per hour to Kilometers per hour'
];

// Mapping for input and output units
final Map<String, String> _inputUnits = {
  'Meters to Kilometers': 'meters',
  'Kilometers to Meters': 'kilometers',
  'Grams to Kilograms': 'grams',
  'Kilograms to Grams': 'kilograms',
  'Celsius to Fahrenheit': 'Celsius',
  'Fahrenheit to Celsius': 'Fahrenheit',
  'Square Meters to Hectares': 'square meters',
  'Hectares to Square Meters': 'hectares',
  'Miles to Kilometers': 'miles',
  'Kilometers to Miles': 'kilometers',
  'Pounds to Kilograms': 'pounds',
  'Kilograms to Pounds': 'kilograms',
  'Liters to Milliliters': 'liters',
  'Milliliters to Liters': 'milliliters',
  'Kilometers per hour to Miles per hour': 'km/h',
  'Miles per hour to Kilometers per hour': 'mph',
};

final Map<String, String> _outputUnits = {
  'Meters to Kilometers': 'kilometers',
  'Kilometers to Meters': 'meters',
  'Grams to Kilograms': 'kilograms',
  'Kilograms to Grams': 'grams',
  'Celsius to Fahrenheit': 'Fahrenheit',
  'Fahrenheit to Celsius': 'Celsius',
  'Square Meters to Hectares': 'hectares',
  'Hectares to Square Meters': 'square meters',
  'Miles to Kilometers': 'kilometers',
  'Kilometers to Miles': 'miles',
  'Pounds to Kilograms': 'kilograms',
  'Kilograms to Pounds': 'pounds',
  'Liters to Milliliters': 'milliliters',
  'Milliliters to Liters': 'liters',
  'Kilometers per hour to Miles per hour': 'mph',
  'Miles per hour to Kilometers per hour': 'km/h',
};

// Input controller
final TextEditingController _inputController = TextEditingController();

// Selected conversion type and result
String _selectedConversion = 'Meters to Kilometers';
double _result = 0.0;

@override

```

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```

Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(
      title: Text(
        'Unit Converter',
        style: TextStyle(
          color: Colors.white,
          fontWeight: FontWeight.bold,
        ),
      ),
      backgroundColor: Colors.deepPurple,
    ),
    body: Padding(
      padding: const EdgeInsets.all(16.0),
      child: Column(
        children: <Widget>[
          // Dropdown for selecting conversion type
          DropdownButton<String>(
            value: _selectedConversion,
            onChanged: (String? newValue) {
              setState(() {
                _selectedConversion = newValue!;
              });
            },
            items: _conversionTypes.map((String value) {
              return DropdownMenuItem<String>(
                value: value,
                child: Text(value),
              );
            }).toList(),
          ),
          SizedBox(height: 20),

          // Input field with dynamic unit label
          Row(
            children: [
              Expanded(
                child: TextField(
                  controller: _inputController,
                  keyboardType: TextInputType.number,
                  decoration: InputDecoration(
                    labelText: 'Enter value',
                  ),
                ),
              ),
              SizedBox(width: 10),
              Text(
                _inputUnits[
                  _selectedConversion]!, // Dynamic unit next to input field
                style: TextStyle(fontSize: 16),
              ),
            ],
          ),
        ],
      ),
    ),
  );
}

```

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```

    ),
    SizedBox(height: 20),

    // Convert button
    ElevatedButton(
      onPressed: _convert,
      child: Text('Convert'),
    ),
    SizedBox(height: 20),

    // Display result with unit
    Text(
      'Result: $_result ${_outputUnits[_selectedConversion]}',
      style: TextStyle(fontSize: 20),
    ),
  ],
),
),
);
}

```

```

// Conversion Logic based on selected type
void _convert() {
  double inputValue = double.tryParse(_inputController.text) ?? 0.0;

  switch (_selectedConversion) {
    case 'Meters to Kilometers':
      _result = _convertMetersToKilometers(inputValue);
      break;
    case 'Kilometers to Meters':
      _result = _convertKilometersToMeters(inputValue);
      break;
    case 'Grams to Kilograms':
      _result = _convertGramsToKilograms(inputValue);
      break;
    case 'Kilograms to Grams':
      _result = _convertKilogramsToGrams(inputValue);
      break;
    case 'Celsius to Fahrenheit':
      _result = _convertCelsiusToFahrenheit(inputValue);
      break;
    case 'Fahrenheit to Celsius':
      _result = _convertFahrenheitToCelsius(inputValue);
      break;
    case 'Square Meters to Hectares':
      _result = _convertSquareMetersToHectares(inputValue);
      break;
    case 'Hectares to Square Meters':
      _result = _convertHectaresToSquareMeters(inputValue);
      break;
    // New conversions
    case 'Miles to Kilometers':

```

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```

    _result = _convertMilesToKilometers(inputValue);
    break;
case 'Kilometers to Miles':
    _result = _convertKilometersToMiles(inputValue);
    break;
case 'Pounds to Kilograms':
    _result = _convertPoundsToKilograms(inputValue);
    break;
case 'Kilograms to Pounds':
    _result = _convertKilogramsToPounds(inputValue);
    break;
case 'Liters to Milliliters':
    _result = _convertLitersToMilliliters(inputValue);
    break;
case 'Milliliters to Liters':
    _result = _convertMillilitersToLiters(inputValue);
    break;
case 'Kilometers per hour to Miles per hour':
    _result = _convertKilometersToMilesPerHour(inputValue);
    break;
case 'Miles per hour to Kilometers per hour':
    _result = _convertMilesToKilometersPerHour(inputValue);
    break;
default:
    _result = 0.0;
}

setState(() {});
}

// Conversion functions for each type
// Length conversions
double _convertMetersToKilometers(double value) {
    return value / 1000; // 1 meter = 0.001 kilometers
}

double _convertKilometersToMeters(double value) {
    return value * 1000; // 1 kilometer = 1000 meters
}

// Weight conversions
double _convertGramsToKilograms(double value) {
    return value / 1000; // 1 gram = 0.001 kilograms
}

double _convertKilogramsToGrams(double value) {
    return value * 1000; // 1 kilogram = 1000 grams
}

// Temperature conversions
double _convertCelsiusToFahrenheit(double value) {
    return (value * 9 / 5) + 32; // (C * 9/5) + 32 = F

```

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```

}

double _convertFahrenheitToCelsius(double value) {
  return (value - 32) * 5 / 9; // (F - 32) * 5/9 = C
}

// Area conversions
double _convertSquareMetersToHectares(double value) {
  return value / 10000; // 1 square meter = 0.0001 hectares
}

double _convertHectaresToSquareMeters(double value) {
  return value * 10000; // 1 hectare = 10000 square meters
}

// New conversion functions
// Length: Miles and Kilometers
double _convertMilesToKilometers(double value) {
  return value * 1.60934; // 1 mile = 1.60934 kilometers
}

double _convertKilometersToMiles(double value) {
  return value / 1.60934; // 1 kilometer = 0.621371 miles
}

// Weight: Pounds and Kilograms
double _convertPoundsToKilograms(double value) {
  return value * 0.453592; // 1 pound = 0.453592 kilograms
}

double _convertKilogramsToPounds(double value) {
  return value / 0.453592; // 1 kilogram = 2.20462 pounds
}

// Volume: Liters and Milliliters
double _convertLitersToMilliliters(double value) {
  return value * 1000; // 1 liter = 1000 milliliters
}

double _convertMillilitersToLiters(double value) {
  return value / 1000; // 1 milliliter = 0.001 liters
}

// Speed: Kilometers per hour and Miles per hour
double _convertKilometersToMilesPerHour(double value) {
  return value * 0.621371; // 1 km/h = 0.621371 mph
}

double _convertMilesToKilometersPerHour(double value) {
  return value / 0.621371; // 1 mph = 1.60934 km/h
}
}

```

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Output:-

